Extracting lithium from the Salton Sea

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Outline

1. Context on Salton Sea
2. Scientific macroscopics
3. Microscopic studies

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SALTON SEA
CALIFORNIA

Historical overview

- Formed when Colorado River breached a levee

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- Used to get water due to agricultural runoff
- Now with more rigorous water management due to water stress, sea is drying up

Politics

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- Trump canceled the loan, so progress has been slow

Environment

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- Regular summer die-offs of fish
- Study found that children growing up there are 4x more likely to develop asthma

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Elevator pitch

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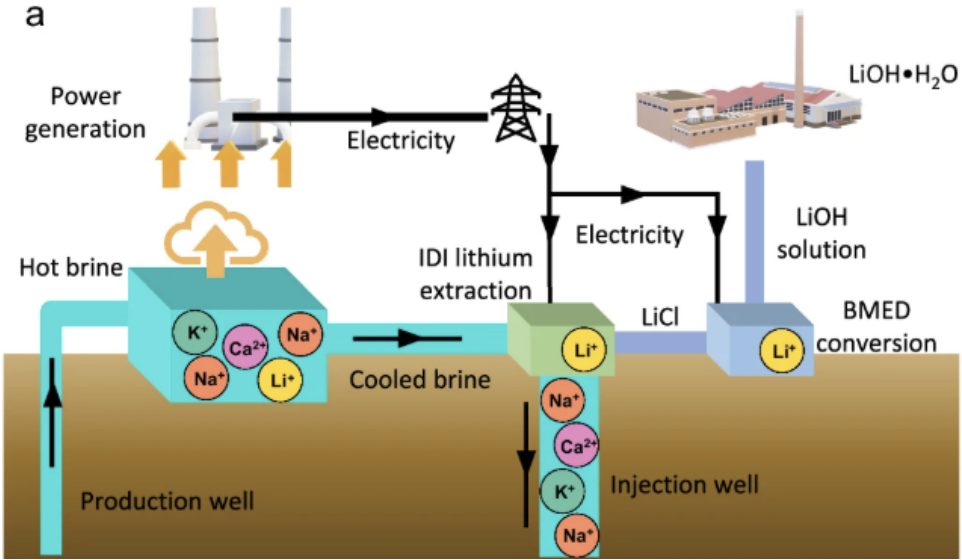
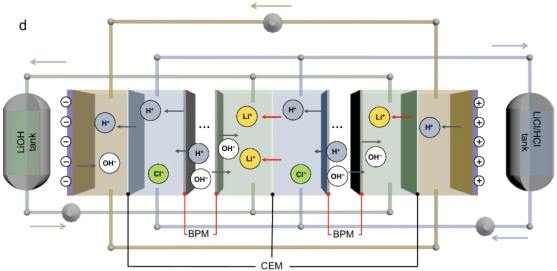
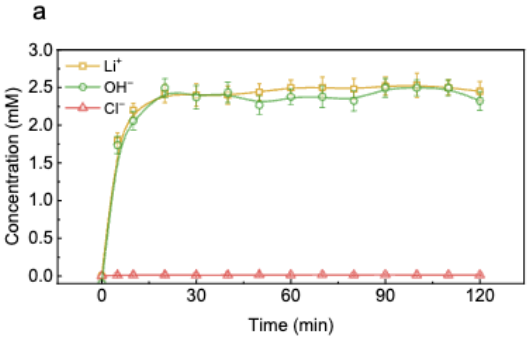
- Electro-driven lithium extraction from brines lessens environmental impact
- Process achieves lithium selectivity over other metal ions.
- Technoeconomic analysis showed lithium production at one-third of market price

Industrial process

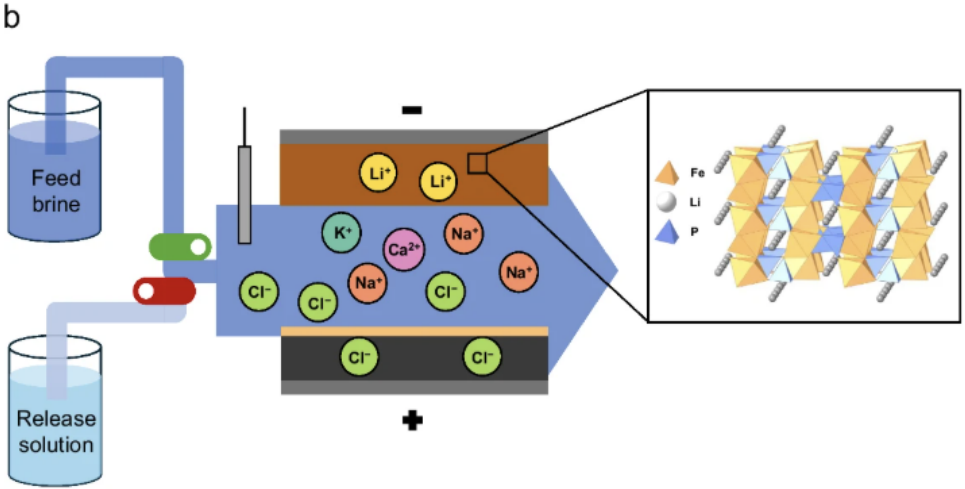


Bipolar membrane electrodialysis





Intercalation deionization



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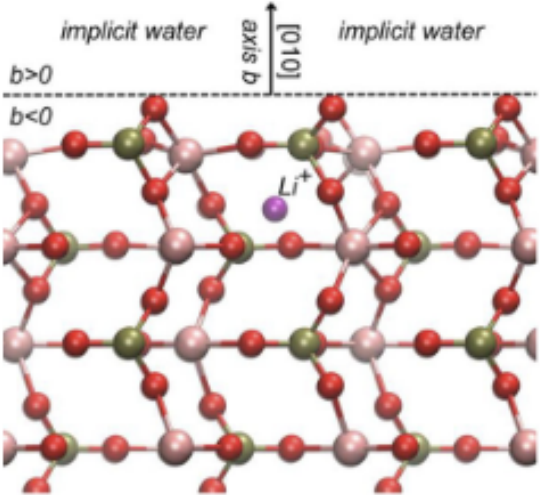
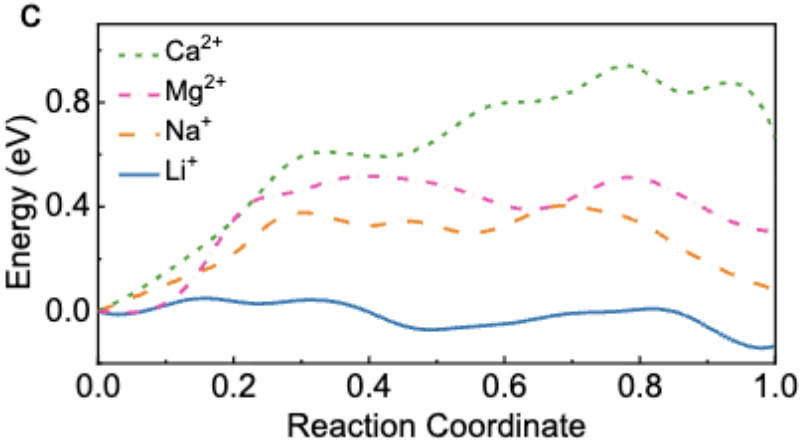
Intercalation material

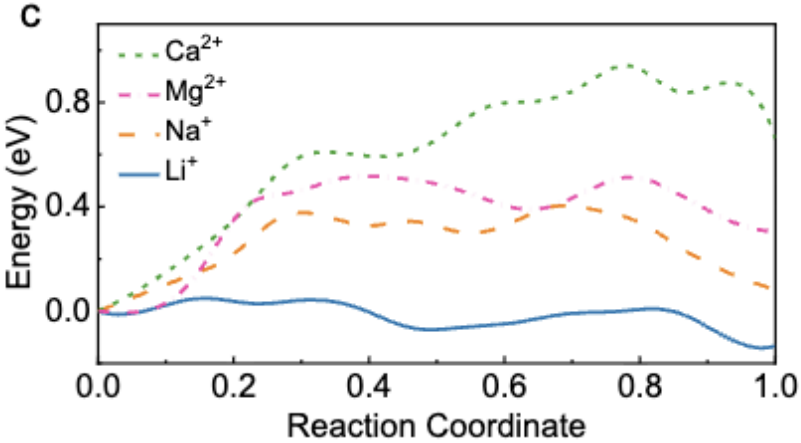
- Molecular dynamics simulations showed selectivity of LiFePO4 toward Li+



DFT study

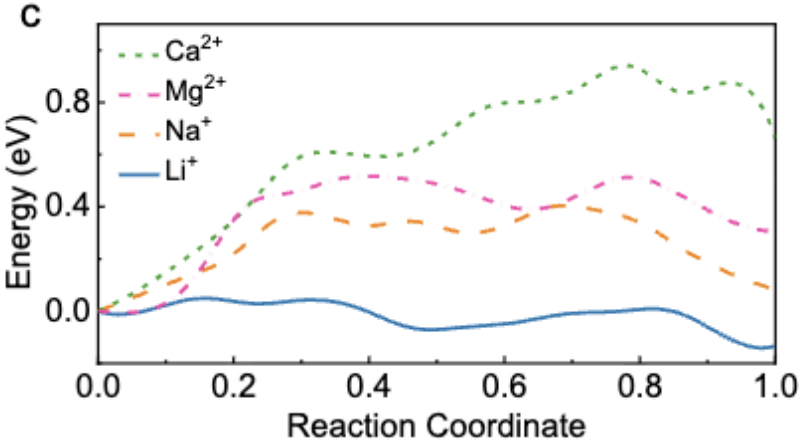

c
Ca2+
Mg2+
Na+
Li+
Energy (eV)
0.0
0.4
0.8
0.0
0.2
0.4
0.6
0.8
1.0
Reaction Coordinate

DFT study



- Smaller ion size means better fit in cage

DFT study



- Smaller ion size means better fit in cage
- Higher valency means greater solvation energy

Questions?